

LankaMart Sales Analysis Project

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AI/ML Ignition Program Batch No.2

1. Import Libraries

These libraries are used for data handling and visualization.

```
#Import numpy, pandas, and matplotlib

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

2. Load Dataset

Load the dataset to start analysis.

```
# Read the dataset
df = pd.read_csv("supermarket_sales_srilanka_1500.csv")

# Display the first few rows of the dataset
df.head()
```

	Invoice_ID	Branch	City	Customer_Type	Gender	Product_Category		
0	SLM-100001	Galle	Galle	Member	Male	Household		
1	SLM-100002	Kandy	Kandy	Member	Male	Household		
2	SLM-100003	Galle	Galle	Normal	Male	Beverages		
3	SLM-100004	Colombo	Colombo	Member	Female	Personal Care		
4	SLM-100005	Kandy	Kandy	Member	Male	Household		
	Unit_Price	Quantity	Total	Date	Time	Payment	Rating	
0	584.33	5	2629.49	2026-02-15	21:51	Cash	6.8	
1	3170.80	3	9131.90	2026-01-14	15:55	Cash	7.7	
2	925.68	6	4998.67	2026-02-16	19:45	Card	6.7	

3	2521.37	6	17851.30	2026-02-17	13:40	Cash	8.0
4	1801.89	6	10378.89	2026-02-16	18:37	eZ Cash	7.9

3. Data Overview

Understand dataset size, structure, and basic statistics.

Check the shape, info, and summary statistics of the dataset

```
df.shape
```

```
df.info()
```

```
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1500 entries, 0 to 1499
```

```
Data columns (total 13 columns):
```

#	Column	Non-Null	Count	Dtype
0	Invoice_ID	1500	non-null	object
1	Branch	1500	non-null	object
2	City	1500	non-null	object
3	Customer_Type	1500	non-null	object
4	Gender	1500	non-null	object
5	Product_Category	1500	non-null	object
6	Unit_Price	1500	non-null	float64
7	Quantity	1500	non-null	int64
8	Total	1500	non-null	float64
9	Date	1500	non-null	object
10	Time	1500	non-null	object
11	Payment	1494	non-null	object
12	Rating	1492	non-null	float64

```
dtypes: float64(3), int64(1), object(9)
```

```
memory usage: 152.5+ KB
```

	Unit_Price	Quantity	Total	Rating
count	1500.000000	1500.000000	1500.000000	1492.000000
mean	1710.407593	3.917333	7126.569253	7.152547
std	1693.475187	1.794244	8727.972507	1.287150
min	121.380000	1.000000	137.050000	3.100000
25%	601.970000	2.000000	1841.367500	6.200000
50%	1264.445000	4.000000	4183.125000	7.200000
75%	2061.880000	5.000000	8921.992500	8.000000
max	9463.720000	8.000000	71312.050000	10.000000

4. Data Cleaning

Fix missing values and clean text fields to ensure accuracy.

```
# Strip leading and trailing whitespace from the "Product_Category" column
df["Product_Category"] = df["Product_Category"].str.strip()

# Fill missing values in the "Payment" column with "Unknown"
df["Payment"] = df["Payment"].fillna("Unknown")

# Fill missing values in the "Rating" column with the mean rating
df["Rating"] = df["Rating"].fillna(df["Rating"].mean())
```

5. Total Sales

This shows the overall revenue generated in the dataset.

```
# Calculate total sales
total_sales = df["Total"].sum()
print("Total Sales:", round(total_sales, 2))

Total Sales: 10689853.88
```

6. Sales by Branch

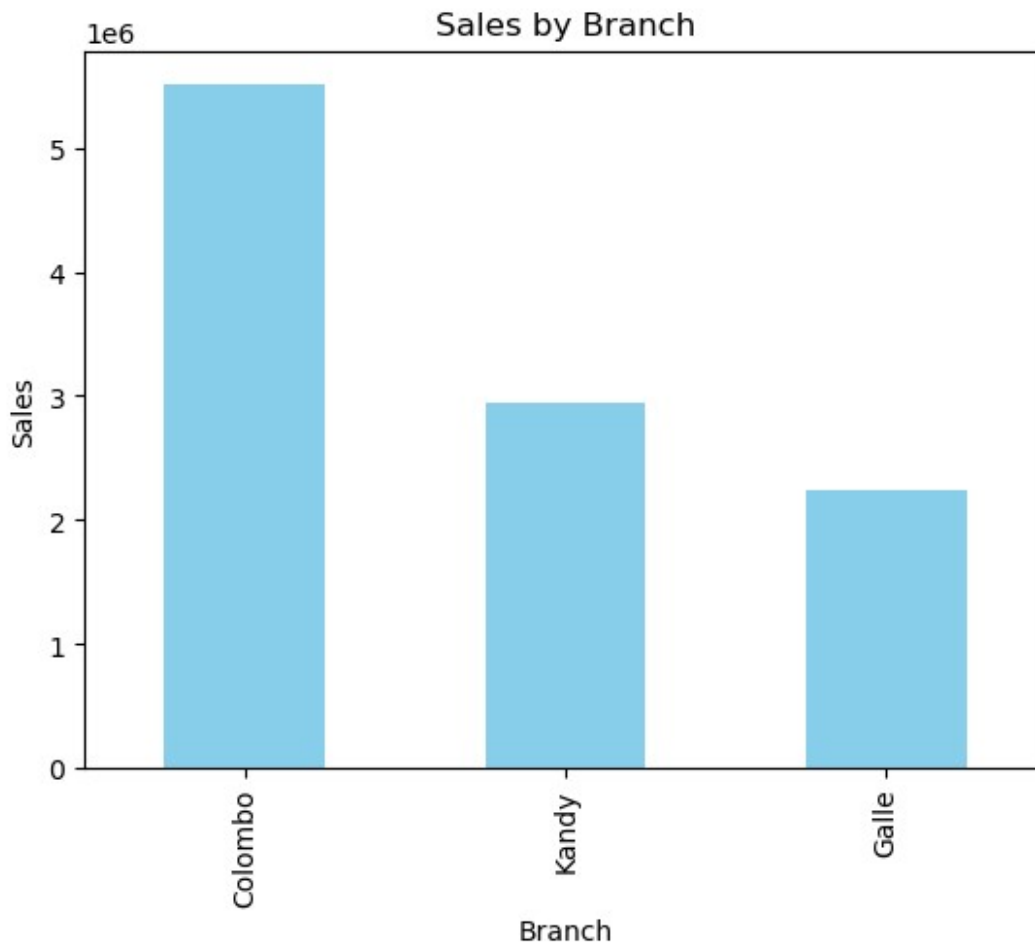
Identify which branch contributes most to total sales.

```
# Calculate total sales by branch using groupby and sort the results in descending order
branch_sales = df.groupby("Branch")
["Total"].sum().sort_values(ascending=False)
branch_sales
```

Branch	Total
Colombo	5507914.09
Kandy	2942373.76
Galle	2239566.03

```
Name: Total, dtype: float64

# Visualize sales by branch using a bar chart
branch_sales.plot(kind="bar", color="skyblue")
plt.title("Sales by Branch")
plt.xlabel("Branch")
plt.ylabel("Sales")
plt.show()
```



Finding: Colombo branch generates the highest revenue, while Galle contributes the least.

7. Sales by Product Category

Analyze which product types bring the most income.

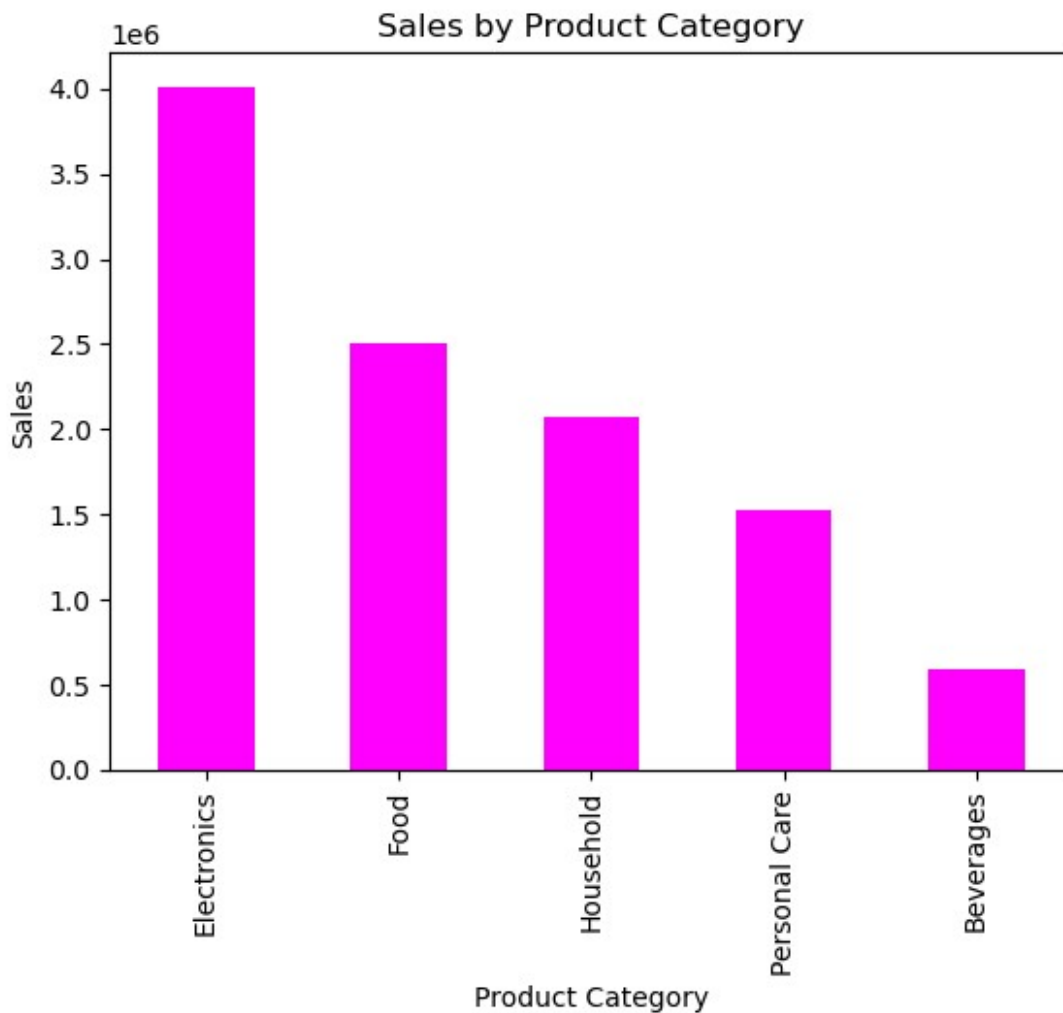
```
# Calculate total sales by product category using groupby and sort the
results in descending order
category_sales = df.groupby("Product_Category")
["Total"].sum().sort_values(ascending=False)
category_sales
```

Product_Category	Total
Electronics	4010868.47
Food	2500763.52
Household	2068873.92
Personal Care	1522994.54
Beverages	586353.43

Name: Total, dtype: float64

```
# Visualize sales by product category using a bar chart
category_sales.plot(kind="bar",color="magenta")
plt.title("Sales by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Sales")
plt.show

<function matplotlib.pyplot.show(close=None, block=None)>
```



Finding: Electronics dominates sales, while Beverages generate the least revenue.

8. Peak Shopping Hours

Analyze customer activity across different hours of the day.

```
# Extract the hour from the "Time" column and create a new "Hour" column
df["Hour"] = pd.to_datetime(df["Time"] , format="%H:%M").dt.hour
```

```
# Calculate total sales by hour using groupby and sort the results in descending order
```

```
hourly_sales = df.groupby("Hour") ["Total"] .sum()  
hourly_sales
```

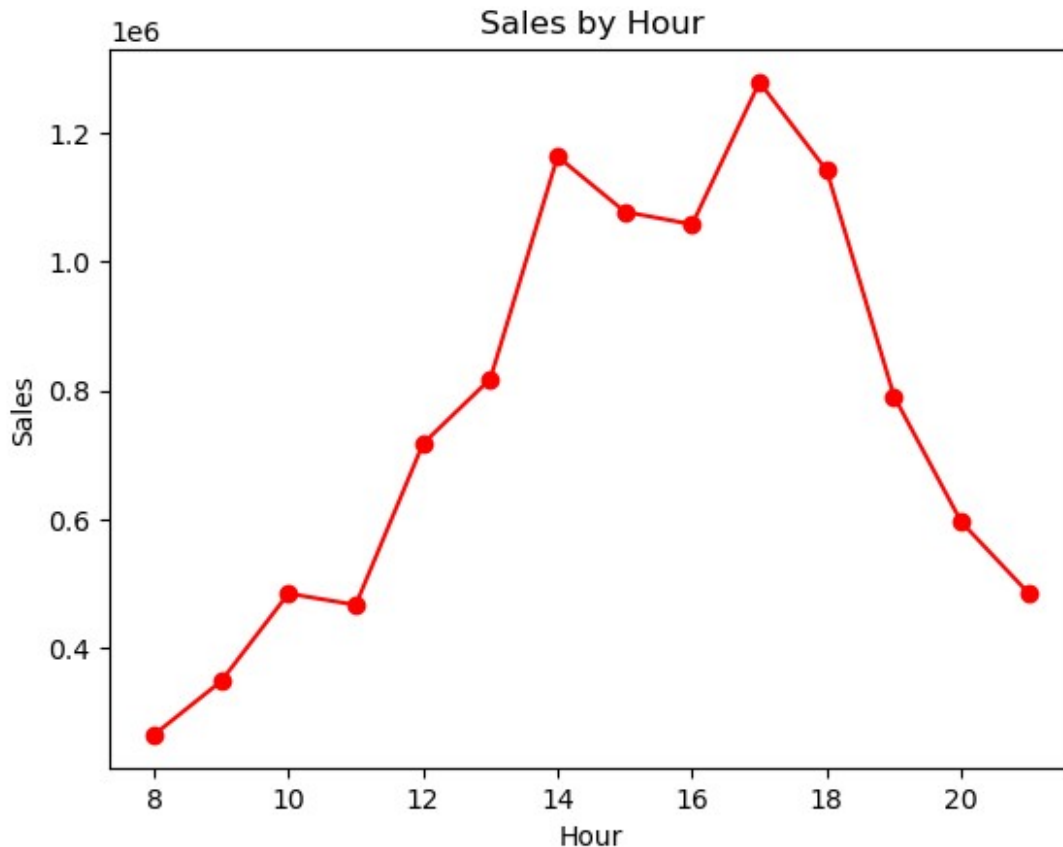
```
Hour
```

```
8      264822.68  
9      348152.57  
10     484427.56  
11     466727.42  
12     717179.35  
13     817923.30  
14    1163419.25  
15    1077544.36  
16    1057953.56  
17    1278969.70  
18    1141597.45  
19     790790.86  
20     595489.87  
21     484855.95
```

```
Name: Total, dtype: float64
```

```
# Visualize sales by hour using a line chart with marke
```

```
hourly_sales.plot(marker="o",color="red")  
plt.title("Sales by Hour")  
plt.xlabel("Hour")  
plt.ylabel("Sales")  
plt.show()
```



Finding: Peak shopping occurs in the afternoon and evening (around 14:00–18:00).

9. Payment Analysis

Understand how customers prefer to pay.

```
# Calculate the number of transactions for each payment method using value_counts
```

```
payment_counts = df ["Payment"].value_counts()  
payment_counts
```

```
Payment  
Cash      683  
Card      523  
eZ Cash   288  
Unknown    6  
Name: count, dtype: int64
```

Finding: cash is the most used payment method, followed by card payments.

10. Customer Satisfaction

Evaluate average ratings per branch.

```
# Calculate the average rating for each branch using groupby and sort
the results in descending order
branch_rating = df.groupby("Branch")
["Rating"].mean().sort_values(ascending=False)
branch_rating
```

```
Branch
Kandy      7.216277
Colombo    7.127929
Galle      7.117763
Name: Rating, dtype: float64
```

Finding: Kandy has slightly higher customer satisfaction compared to other branches.

11. Advanced Analysis

Profit Estimation

Estimate profit based on 20% margin assumption.

```
# Calculate the estimated profit for each transaction (assuming a
profit margin of 20%) and add it as a new column
df["Estimated_Profit"] = df["Total"] * 0.2

# Calculate total estimated profit by branch using groupby and sort
the results in descending order
df.groupby("Branch")
["Estimated_Profit"].sum().sort_values(ascending=False)
```

```
Branch
Colombo    1101582.818
Kandy      588474.752
Galle      447913.206
Name: Estimated_Profit, dtype: float64
```

Member vs Normal Customers

Compare spending behavior.

```
# Calculate total sales, average sales, and transaction count for each
customer type using groupby and agg
df.groupby("Customer_Type") ["Total"].agg(["sum", "mean", "count"])
```

	sum	mean	count
Customer_Type			
Member	5189272.02	7477.337205	694
Normal	5500581.86	6824.543251	806

Finding: Members spend slightly more on average, but Normal Customers contribute higher total sales.

Top Categories per Branch

```
# Calculate total sales by branch and product category using groupby
and sort the results in descending order, then display the top 3
categories for each branch
```

```
category_totals = df.groupby(["Branch" , "Product_Category"])
["Total"].sum().reset_index()
category_totals = category_totals.sort_values(["Branch",
"Total"],ascending=[True, False])
top_products = category_totals.groupby("Branch") .head(3)
top_products
```

	Branch	Product_Category	Total
1	Colombo	Electronics	2186796.11
2	Colombo	Food	1355859.09
3	Colombo	Household	973794.46
6	Galle	Electronics	712621.91
8	Galle	Household	528418.86
7	Galle	Food	511215.98
11	Kandy	Electronics	1111450.45
12	Kandy	Food	633688.45
13	Kandy	Household	566660.60

Peak Hour per Branch

```
# Calculate total sales by branch and hour using groupby and sort the
results in descending order, then display the top 3 hours for each
branch
```

```
grouped = df.groupby(["Branch", "Hour"])["Total"].sum().reset_index()
sorted_df = grouped.sort_values(["Branch", "Total"],
ascending=[True,False])
top_3_per_branch = sorted_df.groupby("Branch").head(3)
```

```
top_3_per_branch
```

	Branch	Hour	Total
6	Colombo	14	716945.79
7	Colombo	15	631204.41
9	Colombo	17	607513.35
23	Galle	17	379704.87
24	Galle	18	235803.86
20	Galle	14	195755.45
36	Kandy	16	340638.16
38	Kandy	18	306539.61
39	Kandy	19	302577.37

12. Key Findings

- Colombo is the strongest performing branch with significantly higher revenue.
- Electronics is the most profitable product category.
- Peak shopping hours are in the afternoon and evening.
- Cash dominates as the preferred payment method.
- Customer satisfaction is consistent , but Kandy performs slightly better.

13. Recommendations

- Increase stock and promotions for high-performing categories like electronics.
- Improve marketing and operational efficiency in lower-performing branches such as Galle.
- Allocate more staff during peak hours to reduce congestion and improve service.
- Enhance digital payment options to encourage non-cash transactions.
- Focus on improving customer service in branches with slightly lower ratings.